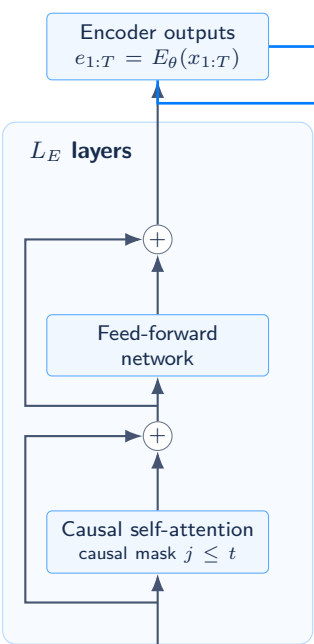


Causal encoder

Known sequence $x_{1:T}$

Causal masking allows parallel encoder prefill across positions.

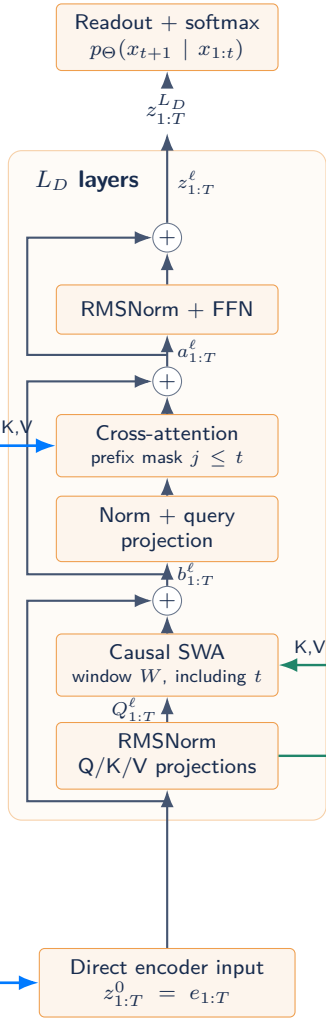


Token embeddings

Known tokens $x_{1:T}$

Decoder w/o feedback

Parallel positions within each layer



Readout + softmax
 $p_{\Theta}(x_{t+1} | x_{1:t})$

$z_{1:T}^{L_D}$

L_D layers

$z_{1:T}^{\ell}$

RMSNorm + FFN

$a_{1:T}^{\ell}$

Cross-attention
prefix mask $j \leq t$

Norm + query
projection

$b_{1:T}^{\ell}$

Causal SWA
window W , including t

$Q_{1:T}^{\ell}$

RMSNorm
Q/K/V projections

Direct encoder input
 $z_{1:T}^0 = e_{1:T}$

One prediction
per position t

Training / prefill
Positions in parallel;
layers in sequence.

Separate KV per layer;
window at position t :
 $j \leq t$ and $t - j < W$

For continuation
save $C_{T,\ell}^{D,\ell}$
last $W - 1$ entries

Layer ℓ SWA KV
 $\{(k_j^{D,\ell}, v_j^{D,\ell})\}_{j=1}^T$

Generation remains
one token at a time.

Same projected KV
read by all decoder layers
($G = 1$ shown)

Attention includes
output projection.
+ denotes a residual addition.
Keys include positional
transformations.